WEEK 1 PROJECT · PYTHON

Employee Attrition Risk Scanner

Business problem: companies lose money when employees leave. This tool flags employees who may be at risk.

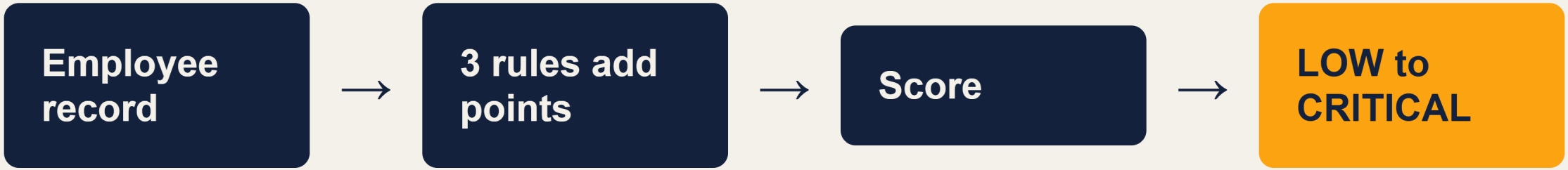
Real data

- Hugging Face: employee-burnout-turnover-prediction-800k
- Synthetic employee records (CC-BY-NC-ND 4.0)
- One record = one employee = one Python dictionary
- Fields used: satisfaction, overtime, career progression

Python concepts learned

- Variables, data types, f-strings
- Lists, strings, sets
- Dictionaries and if / elif / else
- Loops and functions
- try / except error handling

How the system works



Rules: satisfaction below 0.4 (+2), overtime of 10 or more (+2), career progression below 0.4 (+1)

Key code

```
def calculate_risk_score(emp):
    score = 0
    if emp["satisfaction_score"] < 0.4: score += 2
    if emp["overtime_hours"] >= 10: score += 2
    if emp["career_progression_score"] < 0.4: score += 1
    return score
```

A try / except around the loop skips bad records instead of crashing.

Results (1000 employees)

Risk level	Employees	% who left
LOW	599	23.0%
MEDIUM	363	28.9%
HIGH	14	50.0%
CRITICAL	24	37.5%

Higher risk levels left more often than LOW. HIGH and CRITICAL samples are small.

Business value and caveat

- Useful as a quick first look at who may leave
- Rule-based, not machine learning
- Synthetic data, small HIGH and CRITICAL groups
- Not a real employee decision system

What I learned and what is next

- Dictionaries, lists, loops, functions, error handling
- Checking results against real outcomes
- Next: NumPy and Pandas